

Back-and-Forth Adiabatic State Transfer

Round-Trip Dark-State Protocol for a Three-Site Chain

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System

Three coupled sites $q_0 \leftrightarrow q_1 \leftrightarrow q_2$, single-excitation subspace, basis $\{|q_0\rangle, |q_1\rangle, |q_2\rangle\}$:

$$H(t) = \begin{pmatrix} 0 & J_L(t) & 0 \\ J_L(t) & 0 & J_R(t) \\ 0 & J_R(t) & 0 \end{pmatrix}$$

Dark Eigenstate

The zero-energy eigenstate at any instant is

$$|D(t)\rangle = \frac{1}{\sqrt{J_R(t)^2 + J_L(t)^2}} \begin{pmatrix} J_R(t) \\ 0 \\ -J_L(t) \end{pmatrix} = \cos \theta(t) |q_0\rangle - \sin \theta(t) |q_2\rangle, \quad \tan \theta(t) = \frac{J_L(t)}{J_R(t)}.$$

Forward Transfer: $q_0 \rightarrow q_2$ ($0 \leq t \leq T$)

Counter-intuitive pulse order: $J_R(t)$ peaks first (at T_R), $J_L(t)$ peaks later (at $T_L > T_R$).

- $t \approx 0$: $J_L \ll J_R \Rightarrow \theta \approx 0 \Rightarrow |D\rangle \approx |q_0\rangle$
- $t \approx T$: $J_L \gg J_R \Rightarrow \theta \approx \pi/2 \Rightarrow |D\rangle \approx -|q_2\rangle$

Adiabatic following gives

$$|\psi(0)\rangle = |q_0\rangle \implies |\psi(T)\rangle \approx e^{i\phi_f} |q_2\rangle,$$

with ϕ_f a dynamical + geometric phase accumulated along the forward path.

Backward Transfer: $q_2 \rightarrow q_0$ ($T < t \leq 2T$)

Reverse the pulse order in time. Define, for $t \in [T, 2T]$,

$$\tilde{J}_L(t) = J_L(2T - t), \quad \tilde{J}_R(t) = J_R(2T - t).$$

- $t = T$: $\tilde{J}_L \gg \tilde{J}_R \Rightarrow \theta \approx \pi/2 \Rightarrow |D\rangle \approx -|q_2\rangle$
- $t = 2T$: $\tilde{J}_L \ll \tilde{J}_R \Rightarrow \theta \approx 0 \Rightarrow |D\rangle \approx |q_0\rangle$

Starting from $|\psi(T)\rangle \approx e^{i\phi_f} |q_2\rangle$, adiabatic following along the time-reversed dark state gives

$$|\psi(2T)\rangle \approx e^{i(\phi_f + \phi_b)} |q_0\rangle,$$

where ϕ_b is the phase accumulated on the backward path.

Full Round-Trip Evolution

$$H_{\text{round}}(t) = \begin{cases} H(t), & 0 \leq t \leq T \quad (\text{forward}) \\ H(2T - t), & T < t \leq 2T \quad (\text{backward}) \end{cases}$$

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H_{\text{round}}(t) |\psi(t)\rangle, \quad |\psi(0)\rangle = |q_0\rangle.$$

Under adiabatic conditions,

$$|\psi(2T)\rangle \approx e^{i\Phi_{\text{round}}} |q_0\rangle, \quad \Phi_{\text{round}} = \phi_f + \phi_b,$$

and the **round-trip fidelity** is

$$F_{\text{round}} = |\langle q_0 | \psi(2T) \rangle|^2 \approx 1$$

provided the evolution stays adiabatic and leakage to $|q_1\rangle$ (and away from the dark manifold) is negligible.

Observable Quantities

Forward stage ($0 \leq t \leq T$)

$$P_k(t) = |\langle q_k | \psi(t) \rangle|^2, \quad k \in \{0, 1, 2\} \quad (1)$$

$$F_{\text{target}}^{\rightarrow}(t) = |\langle q_2 | \psi(t) \rangle|^2 \quad (2)$$

$$F_{\text{dark}}(t) = |\langle D(t) | \psi(t) \rangle|^2 \quad (3)$$

Backward stage ($T < t \leq 2T$)

$$F_{\text{target}}^{\leftarrow}(t) = |\langle q_0 | \psi(t) \rangle|^2$$

Implementation Notes

1. Run the existing simulation loop for $t \in [0, T]$ unchanged.
2. Continue for $t \in [T, 2T]$, evaluating the couplings at $J_L(2T - t)$ and $J_R(2T - t)$ instead of $J_L(t)$, $J_R(t)$.
3. Plot $P_k(t)$, $F_{\text{target}}^{\rightarrow}(t)$, and $F_{\text{target}}^{\leftarrow}(t)$ over the full interval $[0, 2T]$ to visualize $q_0 \rightarrow q_2 \rightarrow q_0$.
4. A convenient way to reuse the existing `hamiltonian(t)` function is to define `hamiltonian_round(t)` that returns `hamiltonian(t)` for $t \leq T$ and `hamiltonian(2T - t)` for $t > T$.